

Notes: Atkeson and Burstein (JPE, 2010)

Innovation, Firm Dynamics, and International Trade

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1 Big picture

- **Question.** When international trade costs fall, firms change their exit, export, and innovation decisions. Do these heterogeneous firm responses materially change the aggregate welfare gains from trade?
- **Setting.** The paper is a two-country general equilibrium model with heterogeneous firms, fixed and iceberg trade costs, endogenous process innovation inside incumbent firms, and endogenous product innovation through free entry.
- **Core challenge.** A reduction in trade costs has clear *micro* effects: more firms export, less productive firms may exit, and innovative effort can be reallocated toward exporters. The hard question is whether these micro reallocations translate into large *aggregate* productivity and welfare effects.
- **Main conceptual contribution.** The paper distinguishes between two innovation margins: *process innovation* (raising the productivity of an existing firm) and *product innovation* (creating new firms/products). The key contribution is to show that the response of product innovation often offsets the aggregate consequences of the other margins.
- **Main analytical result.** In three tractable special cases, the first-order *indirect* effect of a trade-cost reduction on aggregate productivity is the same as in the older literature with only product innovation. In particular, if the research good is produced only with labor ($\lambda = 1$ in the paper's notation), the indirect effect is zero to a first-order approximation.
- **Economic intuition.** Lower trade costs raise firms' profits. But with free entry, the expected profits from creating a new product must remain zero in equilibrium. Product innovation therefore adjusts so that aggregate productivity moves only as much as is consistent with the free-entry condition.
- **Key micro-to-macro lesson.** A trade-cost reduction can sharply raise the productivity of the *average* firm by reallocating activity toward productive exporters, yet this does not imply a similarly large increase in aggregate productivity or welfare because entry adjusts in the opposite direction.
- **Quantitative result.** In calibrated versions of the model, the offset is not exact once the real interest rate is positive and process innovation is elastic. Still, even then, the change in aggregate productivity is typically much smaller than the change in the productivity of the average firm.
- **Welfare result.** Once transition dynamics are taken into account, welfare gains are close to those in simpler models that abstract from endogenous exit, export, and process innovation, because the extra gains from these margins are either offset in steady state or dissipated by very slow transition dynamics.
- **Takeaway.** Micro evidence showing strong firm-level responses to trade liberalization need not be very informative about aggregate welfare gains from trade.

2 Data and measurement (key objects)

2.1 Why this is a theory paper but still has “measured” objects

- This is not an empirical panel paper. There is no regression data set or instrument.
- Instead, the paper builds a structured dynamic model and then studies two kinds of objects:
 - **model objects** such as firm productivity, export status, free-entry values, and aggregate productivity;
 - **quantitative targets** used to calibrate the model to salient U.S. moments on firm dynamics and trade.
- The relevant “measurement” problem in the paper is therefore: which model objects matter for welfare, how are they aggregated, and which empirical moments pin down the quantitative version of the model?

2.2 Countries, goods, and factors

- There are two countries, home and foreign, in discrete time.
- Each country has:
 - a nontraded final good,
 - a continuum of differentiated intermediate goods that can be traded internationally,
 - and a nontraded **research good** used to pay the costs of operating, exporting, and innovating.
- Households are endowed with labor L and consume the final good.
- The research good is produced using final output and labor, so innovation uses real resources and can reallocate labor away from current production.

Interpretation. The research good is the paper’s bookkeeping device for innovation. It makes process innovation, product innovation, and export fixed costs directly comparable inside one equilibrium system.

2.3 Firms, state variables, and trade costs

Firms are indexed by two state variables:

- z : firm productivity,
- n_x : firm-specific fixed export cost.

A home firm with state $s = (z, n_x)$ produces with labor according to

$$y = e^{z/(r-1)}l, \tag{1}$$

where $r > 1$ is the elasticity of substitution across intermediate goods.

Trade is costly in two ways:

- **iceberg marginal cost** $D \geq 1$,

- **fixed export cost** n_x in units of the research good.

Firms also pay a fixed operating cost n_f each period if they continue.

Interpretation. The paper deliberately combines the Melitz-style export margin with dynamic innovation choices, so the same fall in trade costs can move firms on multiple margins at once.

2.4 Process innovation

Process innovation raises the productivity of an incumbent firm. A continuing firm that spends

$$e^z c(q) \tag{2}$$

units of the research good chooses an innovation intensity $q \in [0, 1]$ and then next period:

- productivity rises from z to $z + \Delta z$ with probability q ,
- or falls from z to $z - \Delta z$ with probability $1 - q$.

The cost function $c(q)$ is increasing and convex.

Interpretation. Process innovation is firm-specific and scales with firm size. Large firms invest more in absolute terms, and the model can generate ongoing firm growth and decline in steady state.

2.5 Product innovation and free entry

Product innovation is entry. Spending n_e units of the research good creates a new firm next period, with initial state drawn from a distribution G .

Free entry requires

$$n_e = \beta \int V(z, n_x) dG(z, n_x), \tag{3}$$

where $V(z, n_x)$ is the expected value of a new firm in units of the research good.

Interpretation. This condition is the central equilibrium force in the paper. Any change in trade costs that raises incumbent profits must be absorbed somewhere else — and the model's answer is usually through a change in product innovation.

2.6 Pricing, export decisions, and aggregate productivity

Monopolistic competition implies constant markups. Given the profit constant P_d , export status is determined by a cutoff rule:

$$x(z, n_x) = 1 \quad \text{iff} \quad P_d D^{1-r} e^z \geq n_x. \tag{4}$$

Define two productivity indices over the steady-state distribution of firms:

$$Z_d = \sum_{n_x} \int [1 - x(z, n_x)] e^z \widetilde{M}(z, n_x) dz, \tag{5}$$

$$Z_x = \sum_{n_x} \int x(z, n_x) e^z \widetilde{M}(z, n_x) dz. \tag{6}$$

The paper’s ideal measure of aggregate productivity is

$$Z = \{M_e [Z_d + (1 + D^{1-r})Z_x]\}^{1/(r-1)}, \quad (7)$$

where M_e is product innovation (the mass of entering firms).

Interpretation. Aggregate productivity depends on both *who survives and exports* and *how many products exist*. This is why average-firm productivity can move very differently from aggregate productivity.

2.7 Aggregate labor allocation and key sufficient statistic

The ratio of total variable profits to total research expenditures is denoted by

$$\zeta = \frac{P_d [Z_d + Z_x(1 + D^{1-r})]}{U}, \quad (8)$$

where U is average research-good expenditure per entering firm.

Two steady-state facts matter a lot:

- the labor allocated to research, L_r , and the ratio C/Y are functions of ζ ;
- as $\beta \rightarrow 1$ (the real interest rate approaches zero), $\zeta \rightarrow 1$, so both L_r and C/Y become independent of trade costs.

Interpretation. This is why the algebra becomes especially sharp in the low-interest-rate special cases.

2.8 Calibration targets and experiments

The quantitative model is disciplined by U.S. firm dynamics and trade moments. Table 1 shows the main targets:

- annual standard deviation of employment growth for large firms: 0.25,
- annual employment-weighted exit rate for firms with more than 500 employees: 0.0055,
- employment-based right-tail coefficient for firms from 1,000 to 5,000 employees: -0.20 ,
- exports/gross output: 0.075,
- employment share of exporters: about 0.40.

Other benchmark choices include:

- $r = 5$,
- $\lambda \in \{1, 0.5\}$ for the labor share in research-good production,
- annualized interest rate either 0 or 0.05,
- process-innovation curvature parameter $b \in \{1200, 30, 10\}$.

Interpretation. High b means process innovation is nearly inelastic; low b means it responds strongly to the improved incentives of exporters after trade costs fall.

3 Models and empirical specifications (all equations + interpretation)

3.1 Static profits and the export cutoff

The final-good sector has CES demand, so firms charge constant markups over marginal cost. Variable profits at home are proportional to $P_d e^z$, and export profits are proportional to

$$P_x e^z = P_d D^{1-r} e^z. \quad (9)$$

Total static profits are

$$\Pi(z, n_x) = P_d e^z + \max\{P_d D^{1-r} e^z - n_x, 0\}. \quad (10)$$

This yields a sharp export threshold:

$$x(z, n_x) = 1 \iff P_d D^{1-r} e^z \geq n_x. \quad (11)$$

Interpretation. Lower trade costs raise export profits disproportionately for high-productivity firms, so trade liberalization reallocates export status toward them.

3.2 Bellman equation and endogenous exit

A firm chooses whether to continue operating and how intensively to innovate. The continuation value solves

$$V^o(z, n_x) = \max_{q \in [0,1]} \left\{ \Pi(z, n_x) - c^z c(q) - n_f + \beta(1-d) \sum_{n'_x} [qV(z + \Delta z, n'_x) + (1-q)V(z - \Delta z, n'_x)] G(n'_x | n_x) \right\}. \quad (12)$$

The firm exits if this continuation value is negative:

$$V(z, n_x) = \max\{0, V^o(z, n_x)\}. \quad (13)$$

Because $V^o(z, n_x)$ is increasing in z , exit is also a cutoff rule in productivity.

Interpretation. Trade-cost reductions can affect exit in two ways: directly through export profitability and indirectly through the innovation incentives of productive firms.

3.3 Free entry and the recursive solution of the steady state

The paper's recursive insight is:

- free entry pins down the variable-profit constant P_d ,
- that determines exit, export, and process innovation,
- which in turn determine Z_d , Z_x , and research expenditures U ,

- and finally product innovation M_e adjusts to satisfy the remaining steady-state equilibrium conditions.

The main free-entry discipline is that the expected value of a new product must equal its entry cost. That is why product innovation absorbs changes in the profitability of incumbent firms.

3.4 Aggregate productivity, output, and labor

Output and productivity are tied together through

$$Y = Z(L - L_r), \quad (14)$$

$$Z = \{M_e [Z_d + (1 + D^{1-r})Z_x]\}^{1/(r-1)}. \quad (15)$$

A useful equilibrium relationship is

$$d \log P_d = (2 - r - \lambda) d \log Z + d \log(L - L_r). \quad (16)$$

Interpretation. The free-entry condition constrains how much aggregate productivity can rise because higher productivity changes real wages, output, and therefore variable profits. The model does not let every margin move independently.

3.5 Direct versus indirect effects of lower trade costs

The paper decomposes the change in aggregate productivity from a change in the iceberg trade cost D into:

- a **direct effect**: productivity rises mechanically because fewer resources are lost in shipping goods internationally;
- an **indirect effect**: firms change exit, export, process innovation, and product innovation decisions.

Formally,

$$d \log Z = \underbrace{-s_x d \log D}_{\text{direct effect}} + \underbrace{\left[\text{change in average-firm productivity} + \frac{d \log M_e}{r - 1} \right]}_{\text{indirect effect}}. \quad (17)$$

In the analytically tractable special cases, the ratio of indirect to direct effects is

$$\frac{\text{Indirect Effect}}{\text{Direct Effect}} = \frac{1 - \lambda}{r + \lambda - 2}. \quad (18)$$

Two implications follow immediately:

- if $\lambda = 1$, the indirect effect is zero to first order;
- if $\lambda < 1$, the indirect effect is positive but has the same first-order form across very different micro structures.

3.6 Special case 1: all firms export

Set fixed export costs to zero, $n_x = 0$. Then all firms export, and the paper proves:

- process innovation does *not* change when trade costs change,
- the indirect effect works entirely through product innovation,
- the aggregate labor allocation is unchanged,
- and the direct/indirect formula above holds exactly to first order.

Interpretation. If all firms export, trade liberalization raises the profits of every firm proportionally. Free entry then forces product innovation to adjust so that incumbents' greater profit opportunities do not translate into a large extra welfare gain.

3.7 Special case 2: no productivity dynamics after entry

Set $\Delta z = 0$. Then firms do not innovate on the process margin after entry, and the model collapses to a Melitz-style environment with fixed export costs.

The paper shows that the first-order indirect effect is again exactly the same as above when the real interest rate is close to zero. Endogenous export and exit decisions still move, but because the marginal firms earn zero current profits at the cutoff, their contribution to aggregate productivity is offset by product innovation.

Interpretation. Endogenous selection alone does not create large additional first-order welfare gains from trade.

3.8 Special case 3: exogenous selection with process innovation

The third special case allows process innovation but makes export status effectively exogenous. Firms move across a two-state export-cost Markov process, so exporters have persistently larger markets and therefore stronger incentives to innovate.

The paper proves that exporters innovate more than nonexporters, so a fall in trade costs reallocates process innovation toward exporters. Even so, with a low real interest rate the same first-order formula for the indirect effect still holds.

Interpretation. Even when trade liberalization changes the allocation of innovative effort across firms, free entry and product innovation neutralize most of the welfare consequences.

3.9 Transition dynamics

The paper emphasizes that transition dynamics can be very slow. In the exogenous-selection version, the aggregate dynamics are governed by the same matrix that determines how much more productive old firms are than entering firms.

Interpretation.

- If entering firms quickly resemble the average firm, transition dynamics are fast and process innovation is quantitatively unimportant.

- If the average firm is much more productive than entrants, process innovation matters a lot, but then the transition to the new steady state is slow.

This trade-off becomes crucial in the welfare analysis.

3.10 Quantitative experiments

The paper runs four main experiments:

- **Experiment 1:** zero interest rate, different elasticities of process innovation;
- **Experiment 2:** positive interest rate with inelastic process innovation;
- **Experiment 3:** positive interest rate with elastic process innovation;
- **Experiment 4:** a larger trade-cost reduction under the positive-interest-rate specification.

Across these exercises, the paper asks two questions:

- How much does aggregate productivity move relative to average-firm productivity?
- Do the larger steady-state changes that can arise with elastic process innovation translate into materially larger welfare gains once transitions are included?

4 Identification and instruments (what makes the estimates credible)

4.1 There is no external instrument

This is a theory paper, so there is no external instrument or empirical identification strategy in the causal-inference sense. Credibility instead comes from:

- transparent equilibrium accounting,
- analytically solved special cases,
- comparison across environments that isolate particular margins,
- and a calibrated quantitative model tied to U.S. firm-dynamics moments.

Interpretation. The paper’s question is not “what is the reduced-form effect in the data?” but “what does equilibrium imply once the firm-level margins are linked together consistently?”

4.2 What pins down the main welfare result

The key disciplining force is the free-entry condition. Lower trade costs raise incumbent profits, but new-product creation must again satisfy zero expected value in equilibrium. That requirement forces a combination of:

- higher wages,

- different aggregate output,
- and different product innovation.

This is why the paper keeps returning to the relation between $d \log P_d$, $d \log Z$, and $d \log(L - L_r)$.

Interpretation. The model does not deny big micro responses. It says those micro responses are equilibrium-restricted and therefore do not freely translate into large welfare effects.

4.3 Why large micro reallocations need not imply large welfare effects

A fall in trade costs often causes:

- more exporting,
- less exit among productive firms,
- more process innovation by exporters,
- and a large rise in the productivity of the average firm.

But aggregate productivity is not the productivity of the average incumbent firm. It also depends on the mass of products in the economy. Product innovation adjusts in the opposite direction, offsetting much of the rise in incumbent productivity.

Interpretation. This is the paper’s core macro message: do not confuse selection/reallocation gains among incumbents with welfare-relevant aggregate productivity gains.

4.4 When the offset is not exact

The exact first-order offset result relies on strong assumptions. The paper shows where it breaks:

- when the real interest rate is positive,
- when process innovation is elastic,
- and when entering firms are small relative to incumbents, so the hybrid export share relevant for discounted profits is much smaller than the actual export share.

In those cases, the indirect effect can be negative or positive depending on how process and product innovation interact.

Interpretation. The paper is not claiming that the offset is universally exact. It is claiming that even where the offset weakens, the extra welfare effect is still surprisingly small.

4.5 Why welfare remains close to the benchmark

The positive-interest-rate, elastic-process-innovation case produces larger steady-state output responses. But the welfare effects stay near the benchmark because the transition is slow.

When entrants are small relative to the average firm, the model implies that the new higher export/output ratio takes decades — in some cases more than 100 years — to materialize. The present

value of these gains is therefore limited.

Interpretation. Faster micro reallocation does not mean faster welfare improvement; in this model, the opposite can be true.

4.6 What the paper can and cannot distinguish

What the paper does well:

- separates process from product innovation conceptually,
- shows which results are purely first-order and which are quantitative,
- and clarifies why micro firm responses need not map into large welfare gains.

What it does not fully do:

- it does not map the model directly to measured productivity in plant data without extra assumptions,
- it abstracts from variable markups and strategic interaction,
- it abstracts from multiproduct firms,
- and it leaves spillover-driven endogenous growth largely for future work.

4.7 Relation to the literature

The paper nests and connects several classic models:

- **Krugman (1980):** all firms export, only product-innovation logic matters;
- **Melitz (2003):** heterogeneous firms with export selection but no post-entry productivity dynamics;
- **Hopenhayn (1992) / Luttmer (2007b):** firm dynamics with exogenous productivity shocks when process innovation is inelastic;
- **Eaton and Kortum (2001):** no change in innovative effort when all firms export;
- **Arkolakis et al.:** welfare gains can often be summarized by trade shares and elasticities rather than by the full richness of firm behavior.

Interpretation. The paper's main contribution is to show that adding richer firm dynamics changes the micro margins a lot more than it changes welfare.

5 Tables and figures

5.1 Table 1: model parameterization

Read.

- The table lists the benchmark parameters and empirical targets for the quantitative model.
- The calibration matches firm-size dynamics, exit, the right tail of the firm-size distribution, the export share of output, and the employment share of exporters.
- The paper varies the curvature of process-innovation costs: $b = 1200$ (high, nearly inelastic), $b = 30$ (moderate), and $b = 10$ (low, elastic).

Economic message. The quantitative section is disciplined by moments that speak directly to the firm dynamics the model is meant to capture, rather than by arbitrary parameter choices.

TABLE 1
MODEL PARAMETERIZATION

		CURVATURE OF PROCESS INNOVATION COST FUNCTION		
	U.S. DATA	High $b = 1,200$ (1)	Moderate $b = 30$ (2)	Low $b = 10$ (3)
Calibrated parameters:				
Exogenous exit rate, δ (annualized)		.005	.005	.005
Process innovation step size, Δ_i (annualized)		.25	.25	.25
Level of process innovation cost function, h (or employment-based right-tail coefficient of large firms)		— (−.25)	3.3E−09 (−.25)	.00108 (−.25)
Marginal trade costs, $D^{(1-\rho)}$		.231	.231	.231
Fixed costs of international trade, n_x		1.4	.7	.285
Targets:				
Employment growth rate of large firms (annual standard deviation)	.25	.25	.25	.25
Annual employment-based exit rate, firms with more than 500 employees	.0055	.0055	.0055	.0055
Employment-based right-tail coefficient, firms from 1,000 to 5,000 employees	−.20	−.20		
Exports/gross output (of intermediate goods in model)	.075	.075	.076	.075
Employment share of exporters (production employment in model)	.40	.402	.404	.402

NOTE.—Other parameters: annualized interest rate, $1/\beta$ annualized = 0 and 0.05; share of labor in production of research good, $\lambda = 1$ and 0.5; elasticity of substitution across intermediate goods, $\rho = 5$; fixed entry cost, $n_e = 1$; fixed operation cost, $n_f = 0.1$.

5.2 Table 2: experiment 1 — zero interest rate

Read.

- With $\lambda = 1$, aggregate productivity moves almost exactly with the direct effect: around 0.075.
- At the same time, the productivity of the average firm can move a lot: roughly 1.17 with moderate process elasticity and 3.89 with low curvature.
- Product innovation offsets those changes almost one-for-one: about −1.18 and −3.91.
- When $\lambda = 0.5$, aggregate productivity is only slightly larger than the direct effect, at about 0.086.

Economic message. This is the cleanest demonstration of the paper’s central result. Big reallocations among incumbents do *not* imply big aggregate productivity gains once product innovation adjusts.

TABLE 2
EXPERIMENT 1: EFFECTS OF A SMALL REDUCTION IN MARGINAL TRADE COSTS, WITH
ZERO INTEREST RATE

	RESEARCH GOOD PRODUCED WITH LABOR ONLY ($\lambda = 1$)			RESEARCH GOOD PRODUCED WITH LABOR + GOODS ($\lambda = 0.5$)		
	(1)	(2)	(3)	(4)	(5)	(6)
Parameters:						
Curvature of process in- novation cost func- tion, b	High	Moderate	Low	High	Moderate	Low
Export share, s_x	.075	.075	.075	.075	.075	.075
Hybrid export share, $\tilde{s}_x$	.075	.075	.075	.075	.075	.075
Elasticity of aggregate vari- ables across steady states*						
Constant on variable profits, Π_d	-.300	-.301	-.303	-.300	-.302	-.303
Aggregate productivity, Z	.075	.075	.076	.086	.086	.086
Direct effect	.075	.075	.075	.075	.075	.075
Productivity of the av- erage firm	.008	1.174	3.885	.008	1.175	3.885
Product innovation	-.008	-1.176	-3.908	.003	-1.166	-3.897
Aggregate production la- bor, $L - L_r$	.000	.000	.000	.000	.000	.000
Output, Y	.075	.075	.076	.086	.086	.086
Consumption, C	.075	.075	.076	.086	.086	.086
Ratio of indirect/direct effects:						
Numerical	.00	.00	.01	.14	.15	.15
Theoretical	.00	.00	.00	.14	.14	.14

5.3 Table 3: experiment 2 — positive interest rate and inelastic process innovation

Read.

- For the “small entering firms” parameterization, the hybrid export share is tiny ($\tilde{s}_x \approx 0.004$), much smaller than the actual export share.
- Aggregate productivity rises by only about 0.010 when $\lambda = 1$ and 0.008 when $\lambda = 0.5$.
- The indirect effect is negative in this case because the decline in product innovation more than offsets the gain in average-firm productivity.
- For the “large entering firms” parameterization, results move back toward the analytical benchmark: aggregate productivity changes by about 0.076 or 0.086.

Economic message. Once the interest rate is positive, the entry margin depends on the discounted present value of future export opportunities. That makes the size of entrants relative to incumbents quantitatively important.

TABLE 3
EXPERIMENT 2: EFFECTS OF A SMALL REDUCTION IN MARGINAL TRADE COSTS, WITH
POSITIVE INTEREST RATE AND INELASTIC PROCESS INNOVATION

	SMALL ENTERING FIRMS		LARGE ENTERING FIRMS	
	$\lambda = 1$ (1)	$\lambda = 0.5$ (2)	$\lambda = 1$ (3)	$\lambda = 0.5$ (4)
Parameters:				
Curvature of process innovation cost function, b	High	High	High	High
Export share, s_x	.075	.075	.078	.078
Hybrid export share, $\tilde{s}_x$	.004	.004	.075	.075
Elasticity of aggregate variables across steady states*				
Constant on variable profits, Π_d	-.017	-.017	-.301	-.301
Aggregate productivity, Z	.010	.008	.076	.086
Direct effect	.075	.075	.078	.078
Productivity of the average firm	.010	.010	.001	.001
Product innovation	-.076	-.078	-.003	.008
Aggregate production labor, $L - L_e$	.022	.011	.002	.001
Output, Y	.032	.019	.077	.087
Consumption, C	.032	.029	.077	.088
Ratio of indirect/direct effects:				
Numerical	-.87	-.89	-.03	.11
Theoretical	-.88	-.90	-.03	.11
Welfare	.076	.073	.078	.087
Welfare in benchmark (all firms export, exogenous exit)	.075	.077	.075	.077

5.4 Table 4 and Figure 1: experiment 3 — positive interest rate with elastic process innovation

Read.

- With elastic process innovation and small entrants, aggregate productivity responds more: about 0.037 with moderate curvature and 0.095 with low curvature when $\lambda = 1$.
- But the productivity of the average firm still moves much more: 0.623 and 2.663, with almost equally large offsetting movements in product innovation.
- Output rises much more than in experiment 2, but welfare remains around 0.076 to 0.078, close to the benchmark.
- Figure 1 shows why: the transition in exports/output is extremely slow when entrants are small, especially with low curvature. It can take more than a century to reach the new steady state.

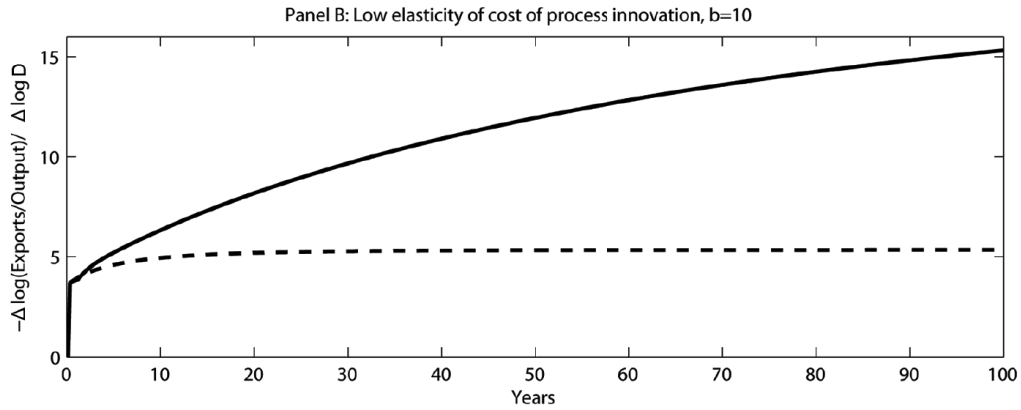
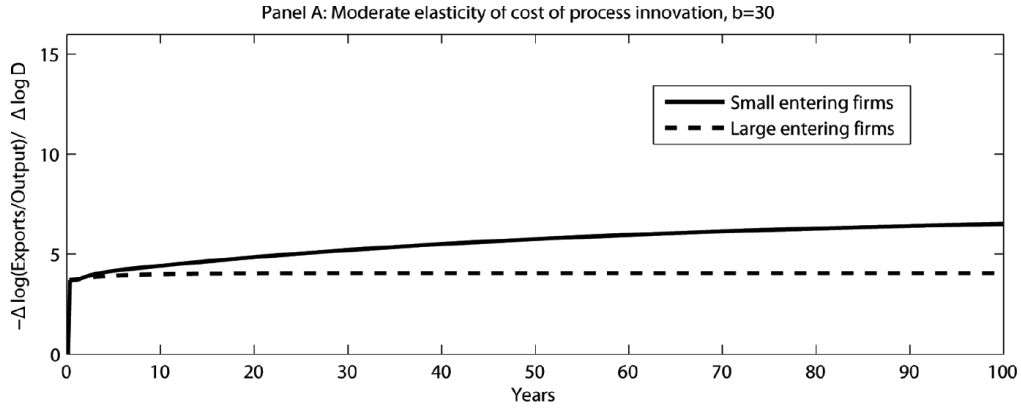
Economic message. This is the paper's most important quantitative qualification. The offset need not be exact, but even when it breaks, the extra welfare gain is modest because the transition is so slow.

TABLE 4
EXPERIMENT 3: EFFECTS OF A SMALL REDUCTION IN MARGINAL TRADE COSTS, WITH
POSITIVE INTEREST RATE AND ELASTIC PROCESS INNOVATION

	SMALL ENTERING FIRMS				LARGE ENTERING FIRMS	
	$\lambda = 1$		$\lambda = 0.5$		$\lambda = 1$	
	(1)	(2)	(3)	(4)	(5)	(6)
Parameters:						
Curvature of process innovation cost function, b	Moderate	Low	Moderate	Low	Moderate	Low
Export share, s_x	.076	.075	.076	.075	.075	.077
Hybrid export share, $\tilde{s}_x$	.009	.022	.009	.022	.073	.072
Elasticity of aggregate variables across steady states*						
Constant on variable profits, Π_x	-.035	-.089	-.035	-.089	-.291	-.289
Aggregate productivity, Z	.037	.095	.027	.071	.074	.076
Direct effect	.076	.075	.076	.075	.075	.077
Productivity of the average firm	.623	2.663	.623	2.663	.046	.258
Product innovation	-.663	-2.654	-.673	-2.678	-.047	-.259
Aggregate production labor, $L - L_x$	.112	.292	.060	.159	.005	.015
Output, Y	.148	.387	.087	.230	.079	.091
Consumption, C	.148	.387	.142	.384	.079	.091
Ratio of indirect/direct effects, numerical	-.52	.26	-.64	-.06	-.02	-.01
Welfare	.076	.076	.076	.078	.075	.077
Welfare in benchmark (all firms export, exogenous exit)	.075	.075	.077	.077	.075	.075

* The negative of (log change in variable/log change in D).

examining the effects of a change in marginal trade costs on aggregate



5.5 Table 5: experiment 4 — a larger trade-cost reduction

Read.

- The paper repeats the analysis for a much larger reduction in trade costs.
- Export shares rise substantially in steady state: from about 7.5% to as much as 16.7% in the low-curvature cases.
- Aggregate productivity rises more than in the small-change exercises — for example to 0.168 when $\lambda = 1$ and process innovation is highly elastic — but the increase in average-firm productivity is still an order of magnitude larger.
- Welfare gains remain close to the benchmark. For example, with $\lambda = 1$ and low curvature, welfare is 0.088 versus 0.080 in the benchmark model.

Economic message. Even large trade-cost changes do not overturn the paper’s basic message: micro firm responses can be dramatic while aggregate welfare changes remain only modestly above the benchmark.

TABLE 5
EXPERIMENT 4: EFFECTS OF A LARGE REDUCTION IN MARGINAL TRADE COSTS, WITH
POSITIVE INTEREST RATE AND ELASTIC PROCESS INNOVATION

	RESEARCH GOOD PRO- DUCED WITH LABOR ONLY ($\lambda = 1$)			RESEARCH GOOD PRO- DUCED WITH LABOR + GOODS ($\lambda = 0.5$)		
	(1)	(2)	(3)	(4)	(5)	(6)
Parameters:						
Curvature of process innova- tion cost function, b	High	Moderate	Low	High	Moderate	Low
Export share, initial steady state	.075	.076	.075	.075	.076	.075
Export share, new steady state	.088	.100	.167	.088	.100	.167
Elasticity of aggregate variables across steady states*						
Constant on variable profits, Π_d	-.019	-.040	-.114	-.019	-.040	-.114
Aggregate productivity, Z	.008	.041	.168	.008	.030	.119
Direct effect + productivity of the average firm [†]	.108	.862	7.518	.108	.862	7.518
Product innovation	-.099	-.821	-7.350	-.100	-.832	-7.399
Aggregate production labor, $L - L_r$	.015	.125	.557	.008	.067	.303
Output, Y	.023	.166	.725	.015	.097	.423
Consumption, C	.023	.166	.725	.022	.158	.711
Welfare	.082	.083	.088	.080	.083	.092
Welfare in benchmark (all firms export, exogenous exit)	.080	.080	.080	.082	.082	.082

* The negative of (log change in variable/log change in D).

[†] We do not separately report the direct and indirect effects on average productivity because eq. (32) is not very

6 Short summary

- The paper builds a heterogeneous-firm trade model with endogenous exit, export, process innovation, and product innovation.
- The central analytical result is that, in several important special cases, the first-order indirect

effect of lower trade costs on aggregate productivity is the same as in simpler product-innovation models.

- Free entry is the reason: if trade liberalization raises incumbent profits, product innovation must adjust so that the expected value of creating a new product stays at zero.
- As a result, the productivity of the average firm can rise a lot even when aggregate productivity and welfare move only modestly.
- In the quantitative model, positive interest rates and elastic process innovation weaken the exact offset, but the extra welfare gains remain small because transition dynamics are very slow.
- The paper's broader message is that micro evidence on firm responses to trade liberalization should not be read as direct evidence about aggregate welfare gains.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that a reduction in trade costs can strongly affect firm-level exit, export, and process innovation decisions without generating comparably large changes in aggregate productivity or welfare. The reason is that product innovation adjusts endogenously through the free-entry condition and offsets much of the aggregate impact.

7.2 Economic intuition

Trade liberalization changes the profitability of selling abroad and therefore the incentives of heterogeneous firms. Productive exporters expand and innovate more, which raises the productivity of incumbents on average. But if new products can be created freely, those greater incumbent profit opportunities must be offset in equilibrium. Entry adjusts, the mass of products changes, and the aggregate consequences become much smaller than the micro reallocations would suggest.

7.3 Takeaway

The main lesson of Atkeson and Burstein is a cautionary one: strong firm-level responses to trade-cost changes are not enough to infer large aggregate welfare gains. In this class of models, the entry margin and transition dynamics matter so much that the macro gains from trade remain close to the benchmark even when the micro responses look dramatic.